

Cationic Copolymer with Amphiphilic Side Chain and Hydrophobic Siloxane/Polycaprolactone Main Chain for Marine Antifouling Coating

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Cite This: *Langmuir* 2025, 41, 8342–8352



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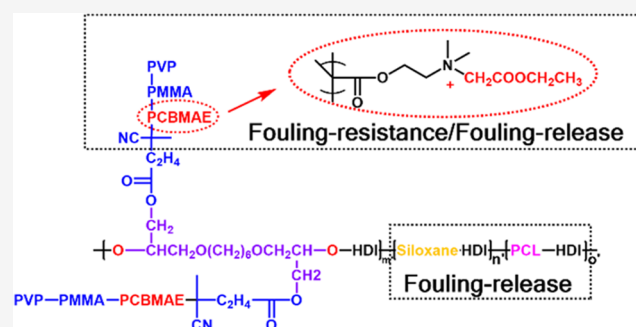
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ABSTRACT: Cationic copolymers formulated with different weight percentages of random copolymerized amphiphilic poly(carboxybetaine methacrylate ester)-poly(methyl methacrylate)-polyvinylpyrrolidone (PCBMAE-PMMA-PVP) side chain and polycaprolactone (PCL)/siloxane main chain were developed via a one-pot continuous reaction for the preparation of marine antifouling coatings. By tuning the weight percentage of the side chain and main chain, a 30% quaternization degree can be achieved without obvious gelation of the cationic copolymer. The hydrolyzable PCBMAE in the side chain and degradable PCL in the main chain led to the self-replenishment of the coating. Moreover, the surface hydrophilicity of the developed coatings was greatly improved owing to the creation of a zwitterionic group during PCBMAE hydrolysis. Results on the weight loss rate of the coating revealed that the amphiphilic side chains, more specifically, the PCBMAE moiety rather than PCL moiety in the hydrophobic main chain, dominated the surface renewal speed. Therefore, the fouling-resistance and fouling-release abilities of the developed cationic copolymer-based coatings were largely promoted compared with the control samples, which only had a main chain. Additionally, experimental results demonstrated that a higher content of amphiphilic side chains correlated with less protein adsorption, but a higher content of siloxane showed more advantages in antibacterial ability. The antibacterial rate of the developed coatings against *Escherichia coli* and *Staphylococcus aureus* ranged from 83 to 95%. The marine field test manifested that the optimized coatings can stand the marine biofouling for four months in the thriving growth season of marine microorganisms.



1. INTRODUCTION

Marine biofouling starts with the bacterial adhesion, followed by marine microorganisms attaching to the surface of underwater facilities.^{1,2} The accumulation of marine organisms not only affects the underwater facility but also causes serious economic loss and environmental problems.³ Therefore, effective control of bacterial adhesion on the surface of the facility is the primary goal of marine antifouling. Constructing efficient and durable coatings on the surface of the marine facility is the most convenient and economical method to prevent marine biofouling.^{4,5}

To meet the needs of environmental protection and underwater durability, a variety of environmentally friendly antifouling coatings have been developed to apply on the surface of underwater facilities, such as low-surface-energy-based materials,^{6,7} biomimetic microstructured materials,⁸ amphiphilic copolymers,^{9,10} slippery liquid-infused porous surface,^{11,12} antibacterial peptides,¹³ etc.

Among them, siloxane-based materials with high elasticity, low surface energy, and smooth surfaces, typically poly(dimethylsiloxane) (PDMS), are considered as the representative fouling-release coatings. In regard to the weak adhesion

force between the foulants and substrate surfaces, foulants can be removed by the shear flow during navigation or by a simple mechanical cleaning.¹⁴

It was proved that the siloxane coating is capable of preventing the settlement of hard marine foulants (such as adult barnacles and mussels) to the hulls of fast cruising vessels,¹⁵ as the shear flow at a high cruising speed facilitated the removal of the foulants adhered to the surfaces. But, PDMS-based coatings have limited antifouling efficacy in resisting the soft marine foulants, which are mainly composed of bacteria, diatoms, and extracellular polymeric substances even in dynamic condition.^{7,14} For the ships sailing at a slower speed and docked at the bay, the problems of slime buildup and foulant accumulation are even worse.¹⁶ In such circumstances, the service life of siloxane-based coatings is greatly

Received: January 10, 2025

Revised: March 6, 2025

Accepted: March 13, 2025

Published: March 20, 2025



shortened. Moreover, the PDMS has poor mechanical properties and its adhesion to the marine facilities is weak, which further limits its application.

The rapid development of various materials provided an opportunity to improve the mechanical and antifouling properties of siloxane-based coatings. Currently, there are two main directions that researchers pursue. One is to strengthen the antifouling ability via introduction of zwitterionic^{17,18} or amphiphilic polymers,¹⁹ which contain a hydrophilic block such as poly(sulfobetaine methacrylate) (PSBMA), poly(carboxybetaine methacrylate ester) (PCBMAE),^{20,21} poly(ethylene glycol) (PEG),²² poly(*N*-vinylpyrrolidone) (PVP),^{23,24} etc., and a hydrophobic block. The other is the combination of self-degradable materials such as polyurethane (PU),^{25,26} polycaprolactone (PCL),^{27,28} 2-methylene-1,3-dioxepane (MDO),²⁹ etc. to accelerate surface renewal.

For example, zwitterionic amphiphilic copolymers including the hydrophobic PDMS block and hydrophilic poly(sulfobetaine methacrylate) (PSBMA) block were developed via controlled free radical polymerization as the additive for marine antifouling coatings.¹⁷ Amphiphilic additives synthesized by attaching hydroxyl-terminated PEG and PDMS chains on a polyisocyanate were blended with PU to fabricate degradable marine coatings,²² but noticeable spherical microdomains resulted in the coating due to the incompatibility of the additive and coating substrate. PVP was functionalized with triethoxysilane and cross-linked with PDMS to form amphiphilic networks to improve the hydrophilicity of the coating.²⁴ PDMS–PVP amphiphilic coatings worked more efficiently in preventing the adhesion of bacteria and marine organisms. Hyperbranched and cross-linked polymers containing the fouling-release PDMS segment and PEG segment exhibited 60% lower bovine serum albumin adsorption than PDMS.³⁰ The above results demonstrated that the incorporation of amphiphilic copolymers clearly has an influence on the antifouling performance of the copolymer, and the hydrophilic/hydrophobic chain length is critical to acquire the desirable fouling-release results.^{17,22,31} However, physical blending of the zwitterionic/amphiphilic copolymers easily leads to the phase separation of the coating; consequently, the surface roughness changes drastically and the release rate of the copolymer cannot be well-controlled.^{17,26}

More importantly, it was proved that comprehensive coatings based on fouling-resistance and fouling-release mechanisms can outperform the coatings based on one platform.³² Hence, integration of multifunctional materials into one coating is usually employed.³³ For example, silicone elastomer with zwitterionic pendant chains was prepared by grafting a telomer of tertiary carboxybetaine dodecafluoroheptyl ester ethyl acrylate (TCBF) and 3-mercaptopropyltriethoxysilane to the bis-silanol-terminated PDMS.^{18,34} The poly(carboxybetaine) analogue hydrolyzed and generated zwitterions on the surface. The marine bacterial adhesion decreases by 80–90% compared with the unmodified PDMS control. PU with zwitterionic side chains based on a poly(carboxybetaine) ester analogue was developed for marine coatings.²⁰ Tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSA) copolymerized with methyl methacrylate (MMA) was developed for marine coatings, as a result of the self-renewal of silyl acrylate and the hydrolysis-induced zwitterionic groups of the copolymer.³⁵ PMMA was used to reinforce the adhesion of the coating to the substrates. Multifunctional antifouling

coatings containing cross-linked perfluoropolyether and PCL chains were developed; the low-surface-energy (22 mJ/m²) materials showed excellent tensile strength and long-lasting antifouling performance.²⁸

Owing to the diversified foulant species and the complexity of marine environments, multifunctional antifouling coatings must meet high standards.³⁶ Inspired by the controlled self-renewal speed of PCL in the copolymer via changing its content,³⁴ and the finely tuned surface properties by introducing amphiphilic copolymer in the coating, polymers based on varied ratios of siloxane and PCL bridged with hexamethylene diisocyanate (HDI) as the main chain and random copolymerization of MMA, dimethylaminoethyl methacrylate (PDMAEMA), and PVP as the side chains were designed in this study. The amphiphilic PDMAEMA–PMMA–PVP side chains were subsequently quaternized as PCBMAE–PMMA–PVP. The designed polymer was synthesized via a continuous one-pot reaction. With the degradation of PCL in the main chain and the hydrolysis of the carboxybetaine unit in the side chains, the residual PCL and siloxane residuals may accumulate on coating surfaces but eventually will be eroded by the water.

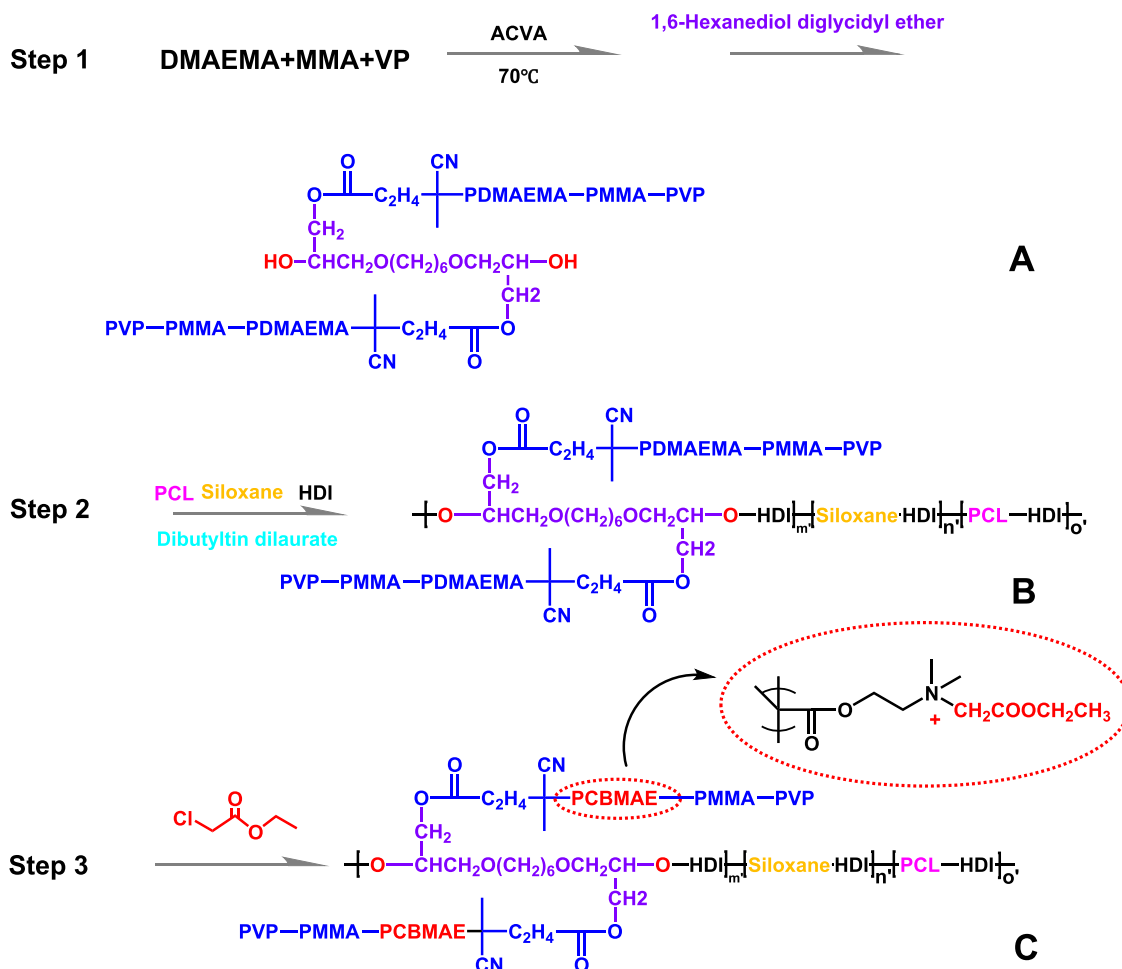
The prepared coatings were anticipated to have enhanced mechanical properties in comparison with pure PDMS, as the main chain was alternately composed of a “soft” segment of siloxane and a “hard” segment of PCL. Meanwhile, hydrolysis-generated zwitterionic groups and PVP of the side chain may endow the materials with enhanced hydrophilicity and antifouling properties. In this study, the weight percentages of the degradable main chain and amphiphilic side chain are important for coating performance, as the weight percentage of PCL is relevant for the weight loss rate of the coating, and the weight percentage of PCBMAE–PMMA–PVP determines the hydrophilicity of the coating. To evaluate the influences of functional domains on the properties of the coatings, different molecular weight ratios of hydrophilic/hydrophobic domains of the copolymer were synthesized and the relationship between structure and function was investigated. It is well known that the settlement of marine fouling organisms relies on the surface and bulk properties of the coating, such as surface free energy, wettability, roughness, elasticity, etc. Hence, the surface physicochemical properties and antifouling performances were all examined.

2. EXPERIMENTAL SECTION

2.1. Materials. Commercially available dimethylaminoethyl methacrylate (DMAEMA), methyl methacrylate (MMA), *N*-vinylpyrrolidone (VP), hexamethylene diisocyanate (HDI), hydroxyl-terminated polycaprolactone (PCL, M_w 1000), hydroxypropyl-terminated siloxane (siloxane, M_w 1000), xylene, dibutyltin dilaurate, 1,6-hexanediol diglycidyl ether, and initiator azobiscyanopentanoic acid (ACVA) were obtained from Aladdin (Shanghai, China). Ethyl chloroacetate and petroleum ether were purchased from Guoyao Chemical Reagent (Tianjin, China). Bovine serum albumin was purchased from Yuanye Biotech (Shanghai, China). The commercially available methacrylate-based CSPC materials containing 40% Cu₂O were used for the marine field test (Kailin Materials, Anhui), and the antifouling agent 4,5-dichloro-2-octyl-isothiazolone (DCOIT) was from Shanghai Jiuding Chemical Co. Ltd. DMAEMA and MMA were passed through a basic alumina column (Al₂O₃) to remove the 4-methoxyphenol inhibitor; the elute was stored at 4 °C for use. Xylene was dried with CaH₂ and stored in molecular sieves.

2.2. Synthesis of Copolymer A with End-Functionalized Hydroxyl Group. The polymer PDMAEMA–PMMA–PVP was synthesized as previously described³⁷ (Scheme 1, step 1). Briefly,

Scheme 1. Synthetic Route of Cationic Copolymers with PCBMAE–PMMA–PVP as Side Chain and Siloxane–HDI–PCL as Main Chain



ACVA (0.28 g, 1 mmol) was first added to a 250 mL three-necked flask and degassed with nitrogen; then, a mixture of the three monomers including DMAEMA (12.56 g, 80 mmol), MMA (6.01 g, 60 mmol), and VP (6.67 g, 60 mmol) was bubbled with nitrogen in a test tube for half an hour. Subsequently, the mixture was introduced into the flask under nitrogen atmosphere, and the reaction was carried out at 69 °C for 3 h to obtain the amphiphilic copolymer PDMAEMA₄₀-PMMA₃₀-PVP₃₀. Its structure was verified via ¹H NMR (400 MHz, Bruker, Germany).

Following the above reaction, xylene (60 g) was added to the system to make the concentration of the amphiphilic copolymer PDMAEMA₄₀-PMMA₃₀-PVP₃₀ 30% (wt %). Then, an equal molar amount of 1,6-hexanediol diglycidyl ether (0.23g, 1 mmol) as that of ACVA was added to the solution. The epoxy ring-opening reaction was proceeded at 80 °C for 6 h to obtain the copolymer A with the end-functionalized hydroxyl group as shown in Scheme 1, step 1.

2.3. Synthesis and Quaternization of Copolymer A. Copolymer A (3 g, 0.12 mmol), PCL (2.56 g, 2.56 mmol), siloxane (2.56 g, 2.56 mmol), HDI (0.88 g, 5.24 mmol), xylene (29 g), and the catalyst dibutyltin dilaurate (0.02 g) were reacted under stirring at 60 °C for 6 h to obtain copolymer B in a round-bottom flask (Scheme 1, step 2).

Following the synthesis procedures described above, ethyl chloroacetate (0.35 g, 2.855 mmol) was added to the mixture in the flask; the quaternization reaction continued at 60 °C for 30 h to obtain copolymer C³⁸ (Scheme 1, step 3). The obtained cationic copolymer C with PCBMAE–PMMA–PVP as the side chain and siloxane–HDI–PCL as the main chain was employed to fabricate coatings. Small amounts of the reaction products from steps 1, 2, and

3 were precipitated in petroleum ether, dried, and subjected to ¹H NMR, respectively.

The monomer feeding ratio described above is listed as coating A in Table 1. A variety of coating components including the side chain/

Table 1. Molecular Weight Ratio of Different Components in the Coatings

	side chain (wt %)	siloxane (wt %)	PCL (wt %)	HDI (wt %)	quaternization degree (%)
coating A	33	28.50	28.50	10.0	30
coating B	33	38.00	19.00	10.0	30
coating C	33	42.75	14.25	10.0	30
coating D	25	32.00	32.00	11.0	30
coating E	25	42.66	21.33	11.0	30
coating F	25	48.00	16.00	11.0	30
control 1		42.80	42.80	14.4	
control 2		57.06	28.53	14.4	
control 3		64.20	21.40	14.4	

main chain of 1:2 (coatings A, B, and C) and 1:3 (coatings D, E, and F) were synthesized to investigate the surface properties of the copolymer-fabricated coatings. Meanwhile, effects of the degradation rate on the antifouling performance were explored by tuning the weight percentage of siloxane/PCL from 1:1 (coating A and D) to 2:1 (coating B and E) to 3:1 (coating C and F). Following the same ratio of siloxane/PCL, three control groups listed as control 1 ~ 3 were prepared (Table 1).

The synthesized polymers were tested by Fourier infrared spectroscopy-attenuated total reflection (FTIR-ATR, Thermo Electron Nicolet 380) in the wavenumber range of 4000–400 cm^{-1} .

2.4. Coating Preparation and Characterization. **2.4.1. Coating Fabrication.** A 30 g coating solution with the weight percentage of 20% cationic copolymer and 80% xylene was spread on the surface of a $20 \times 25 \text{ cm}^2$ poly(tetrafluoroethylene) (PTFE) plate; after naturally drying under room temperature, the film thickness ($\sim 200 \mu\text{m}$) was obtained. The prepared coating was subjected to a film physicochemical test.

2.4.2. Surface Characterization. The coatings were immersed in seawater for hydrolysis and water contact angles on the coating surfaces were tested every 2 days using a goniometer (JGW-360A goniometer, Chengde, China) of the same coating. The Owens–Wendt–Rabel–Kaelble method³⁹ was utilized to measure the contact angle of water and diiodomethane in order to calculate the surface energy of the coating. The values of WCA and surface energy were the average of 5 repetitive tests on different locations of each sample.

Samples for the scanning electron microscopy (SEM) (ZEISS, Germany) test were prepared by placing the coating in 2% NaOH for 3 days to accelerate the hydrolysis. The samples obtained before and after hydrolysis were vacuumed and sprayed with gold for SEM observation.

2.4.3. Surface Renewal Speed. The coating was cut into $2 \times 2 \text{ cm}^2$ pieces and the initial weight of the cut film was measured as W_0 . The film was then placed in 30°C seawater for a time period of 63 days for hydrolysis, and the weight loss was measured every 7 days. Next, the coating was taken out and flushed with water to remove the adhered foulants; after drying under vacuum, the weight of the film was measured as W_1 . The surface renewal rate ($\text{SRS}_t\%$) was then calculated based on the weight loss of the film

$$\text{SRS}_t\% = \frac{W_0 - W_1}{2W_0} \times 100\% \quad (1)$$

Three parallel experiments were conducted on each film and the presented $\text{SRS}_t\%$ value was the average of three tests.

2.4.4. Protein Adsorption Test. Coatings with a size of $2 \times 2 \text{ cm}^2$ were hydrolyzed in 2% NaOH solution for 2 days first and immersed in a BSA solution with a concentration of 1.0 g/L (C_0). After static adsorption for 24 h, coatings were removed from the solution and the concentration of residual solution (C_1) was subjected to ultraviolet–visible absorption test at 280 nm (Beijing Purkinje General Instrument Co., Ltd.). The amount of BSA adsorption on the coatings was calculated following the equation below

$$Q = \frac{(C_0 - C_1) \times V}{S} \times 1000 \quad (2)$$

where Q represents the weight of BSA adsorbed on unit area ($\mu\text{g}/\text{cm}^2$), V (mL) is the volume of the BSA solution, and S represents the area of all of the tested films (cm^2). Each value was the average of at least three measurements of different samples.

2.4.5. Antibacterial Test. *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) were chosen to test the antibacterial ability of the coating.⁴⁰ Briefly, coatings with a size of $1 \times 1 \text{ cm}^2$ were sterilized and incubated in the bacterial solution with a concentration of 10^7 CFU/mL at 37°C for 24 h. The sample was first rinsed with water to remove bacteria that were not adsorbed on the surface and then transferred to 5 mL of PBS followed by sonication to remove the bacteria adhered on the surface. The obtained bacterial solution was diluted and 100 μL was transferred to the Petri dish with the agar medium; the dish was then incubated at 37°C for 16 h. The number of colonies was counted to evaluate the antibacterial rate, which was calculated according to the following equation

$$\text{antibacterial rate}(\%) = \frac{A - B}{A} \times 100\% \quad (3)$$

where A is the average colony number on the control sample, which is the epoxy-coated plate in this study, and B is the colony number on

the surface of the tested coating. Three parallel tests were carried out to obtain the average antibacterial rate of the corresponding coating.

2.4.6. Mechanical Test. The coating was cut into $4 \times 45 \text{ mm}^2$ stripes; the tensile strength and elongation at break of the stripes were then tested via a 10 KN universal material testing machine (US Instron).

2.5. Marine Field Test. A homogeneous solution containing 20% cationic copolymer, 2% DCOIT, and 78% (wt %) xylene was coated on the epoxy plate. After natural drying, the coating was placed in the sea with a depth of 2–3 m underwater at Weihai $37^\circ31'\text{N}$, $122^\circ01'\text{E}$. The tests began on May 15th, 2024 until Oct 15th 2024, during which the coating was taken out every month to observe the foulant adhesion.

3. RESULTS AND DISCUSSION

3.1. Structure Characterization. FTIR test was used to survey the featured spikes of the polymer. The spectra in

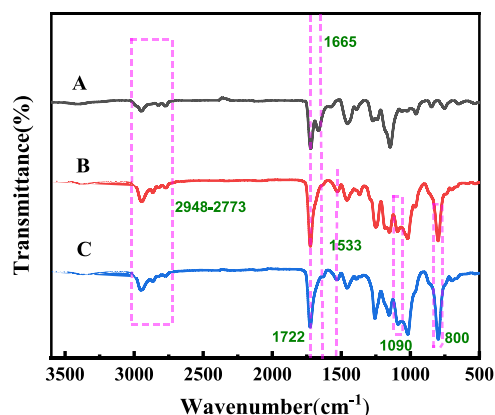


Figure 1. FTIR spectra of copolymer A with amphiphilic PDMAEMA–PMMA–PVP side chain (A), copolymer B with PDMAEMA–PMMA–PVP side chain and siloxane–HDI–PCL main chain (B), and cationic copolymer C with PCBMAE–PMMA–PVP side chain and siloxane–HDI–PCL main chain (C).

Figure 1 displays similar absorption spikes at 2948–2773, 1722, and 1665 cm^{-1} for all three tested polymers as they possessed similar chemical structures. Among them, spikes at 2948–2773 cm^{-1} represented the stretching vibration of $-\text{CH}_3$ and $-\text{CH}_2-$ in DMAEMA and MMA segments.⁴¹ The sharp absorption spike at 1722 cm^{-1} was from the stretching vibration of $\text{C}=\text{O}$. The weak absorption spike at 1665 cm^{-1} , especially in Figure 1B,C, can be assigned to the stretching vibration of the VP ring.⁴² Different from Figure 1A, Figure 1B, C shared absorption peaks at 1533, 800, and 1090 cm^{-1} because they have the same main chain. The spike at 1533 cm^{-1} in Figure 2B,C correlated with the bending vibration of $\text{N}-\text{H}$ resulting from the reaction of the hydroxyl group and isocyanate in the HDI segment. Also, the spike at 800 cm^{-1} attributable to the $\text{Si}-\text{C}$ bending vibration and the spike at 1090 cm^{-1} originating from $\text{Si}-\text{O}-\text{Si}$ asymmetric vibration proved the presence of a siloxane segment.

The ^1H NMR spectrum further validated the detailed structure of the synthesized polymers. The ^1H NMR spectrum of the resulting amphiphilic side chains is shown in Figure 2A, where the peaks at 2.28 ppm for a, 2.56 ppm for b, and 4.06 ppm for c can be ascribed to $-\text{N}-\text{CH}_3$, $-\text{NCH}_2-$, and $-\text{OCH}_2-$ groups in the DMAEMA moiety, respectively. The peak e at 3.59 ppm was from the $-\text{O}-\text{CH}_3$ group in MMA, and peak f at 3.08 ppm can be attributed to the methylene group attached to the tertiary amine on the VP ring. All of

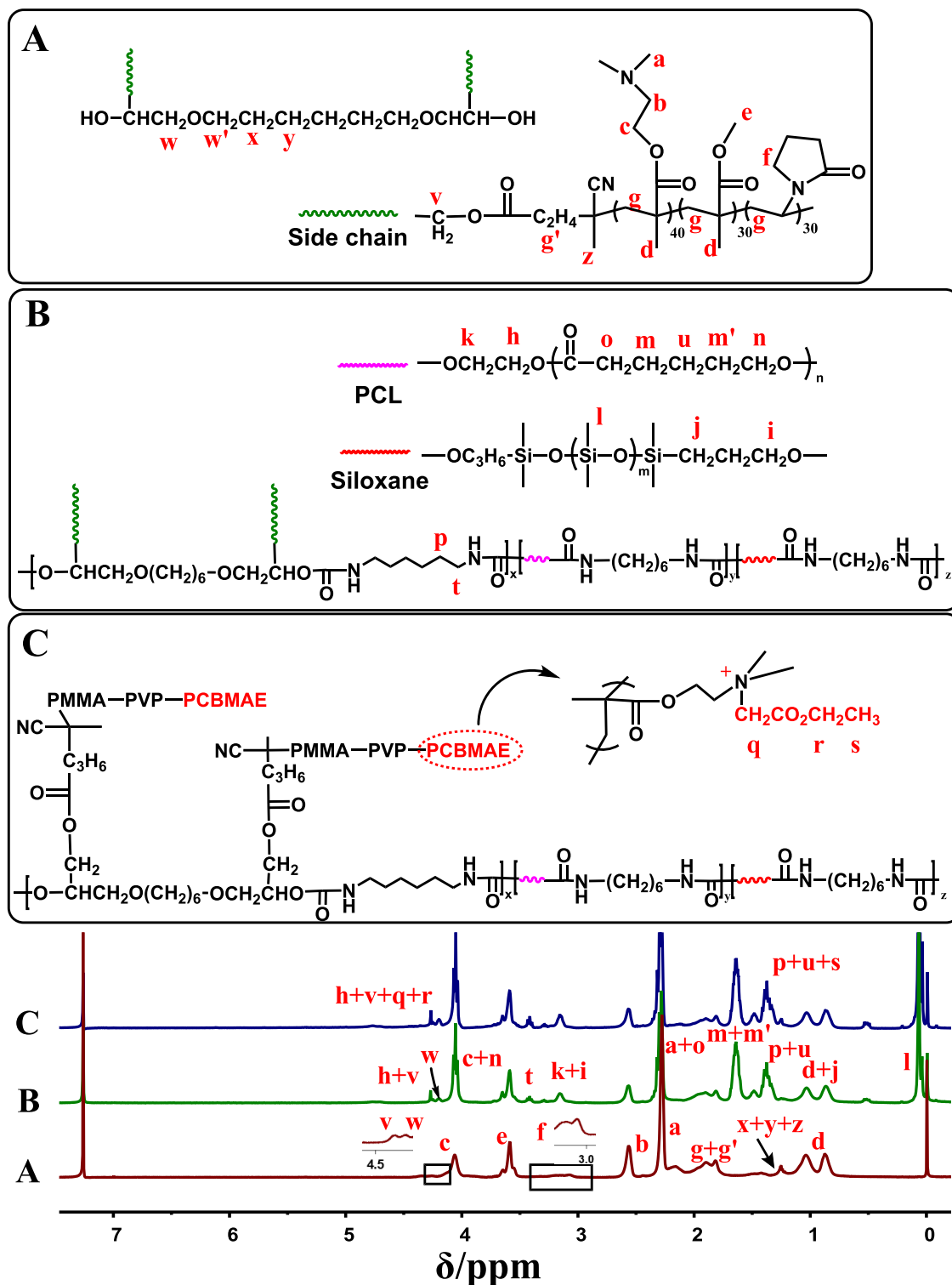


Figure 2. ^1H NMR spectra of copolymer A with amphoteric PDMAEMA–PMMA–PVP side chain (A), copolymer B with amphoteric PDMAEMA–PMMA–PVP side chain and siloxane–HDI–PCL main chain (B), and cationic copolymer C with PCBMAE–PMMA–PVP side chain and siloxane–HDI–PCL main chain (C).

these characteristic peaks indicated the successful synthesis of PDMAEMA–PMMA–PVP. Additionally, peaks w at 4.20 and v at 4.27 ppm were contributed by $-\text{OCH}_2-$ and $-\text{COOCH}_2-$ groups, respectively, which resulted from the ring-opening reaction between the carboxyl group and epoxy group. All of the above featured peaks evidenced the successful synthesis of polymer A.

After the completion of the reaction between polymer A, siloxane, and PCL, the structure of the obtained product was tested via ^1H NMR. Figure 2B clearly shows that peak l at chemical shift 0.06–0.09 ppm belonged to the Si-CH_3 group in the siloxane segment and the chemical shift at 4.27 ppm corresponded with h of the $-\text{COOCH}_2-$ group in the PCL segment. Other methylene peaks from the $-\text{COO}-$

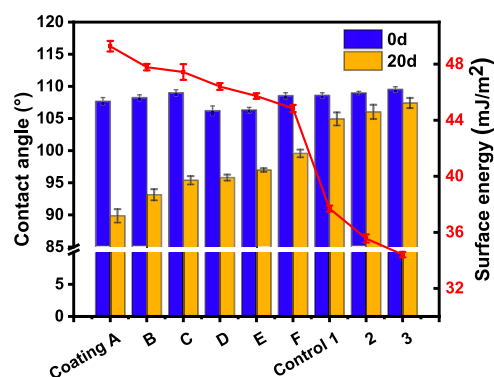


Figure 3. Variations of WCA and surface energy of different coatings after hydrolysis in seawater.

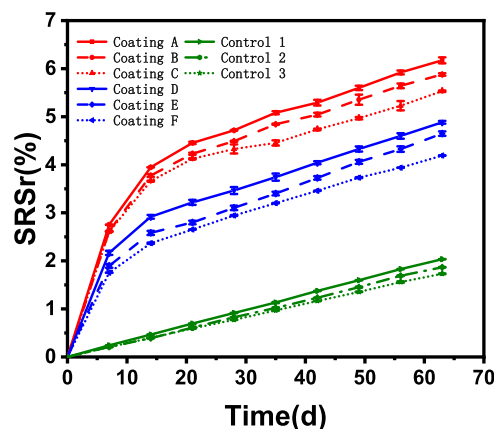


Figure 4. Surface renewal speed of coatings in seawater for 63 days.

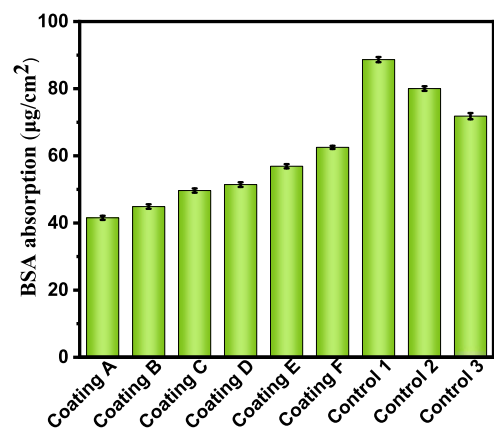


Figure 5. Amount of BSA adsorption on coating surfaces.

$\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ group in the PCL segment, such as m' , k , and u , were found at the chemical shifts of 1.64, 3.15, and 1.37 ppm, respectively. The peak p at 3.41 ppm and peak t at 1.37 ppm can be assigned to the methylene groups of the $-\text{NH}-\text{CH}_2-\text{CH}_2-$ group in the HDI segment, respectively. All of these characteristic peaks confirmed the successful synthesis of the desired copolymer.

After the quaternization reaction, the spectra in Figure 2B,C were similar. Unfortunately, the anticipated signals from the $-\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ group introduced by ethyl chloroacetate and the signals in PCL and HDI overlapped as shown in Figure 2C. To further justify the structure of the reaction product, areas of the characteristic peaks including a at 2.28 ppm and b

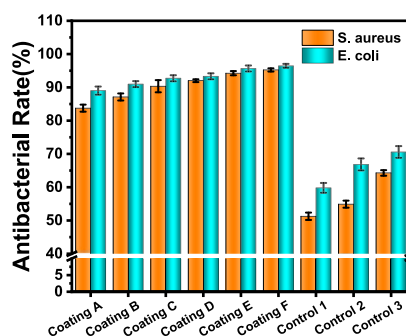
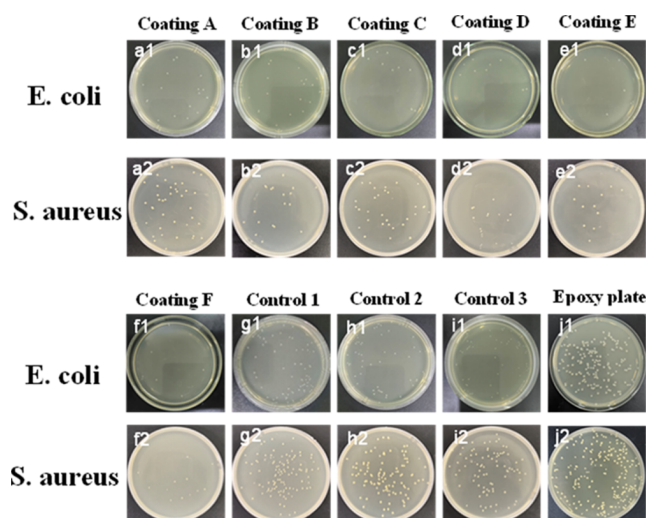


Figure 6. Antimicrobial assay of coatings against *E. coli* (a1–j1) and *S. aureus* (a2–j2), and antibacterial rates of different coatings against *E. coli* and *S. aureus*.

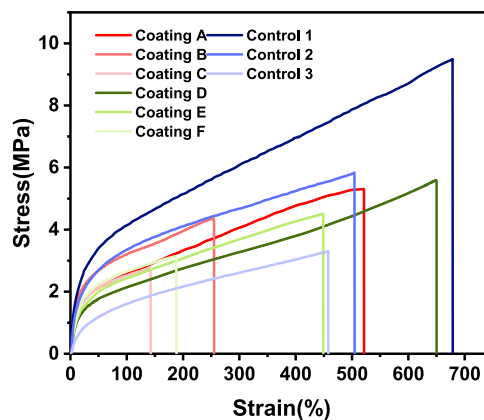
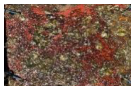
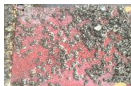


Figure 7. Tensile stress–strain at the break curve of coatings.

at 2.56 ppm for the PDMAEMA block were integrated. Assuming the integral area of $\text{Si}-\text{CH}_3$ (peak l) is 1.00, the peak areas of a at 2.28 ppm and b at 2.56 ppm were 1.17 and 0.26, respectively (Figure S1). After quaternization, the peak areas at 2.28 and 2.56 ppm reduced to 0.69 and 0.15, respectively, as shown in Figure S1. The decreased ^1H NMR signal of PDMAEMA block may have been caused by the self-assembly behavior of the cationic amphiphilic copolymer in the aprotic solvent CDCl_3 ,^{38,43} as the positively charged PCBMAE block may have been encapsulated by the hydrophobic groups and lost its ^1H NMR signal. The weakened ^1H NMR signal of $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$

Table 2. Marine Field Test of the Developed Coatings

	30 d	60 d	90 d	120d	150d
Coating A					
Coating B					
Coating C					
Coating D					
Coating E					
Coating F					
Control 1					
Control 2					
Control 3					
CSPC					
Blank					

N-CH₃ and -NCH₂- associated with PCBMAE block occurred after the quaternization reaction, suggesting the successful implementation of the designed synthetic route (Scheme 1).

3.2. Surface Hydrophilicity and Surface Energy. The surface physicochemical properties were examined by measuring the WCA and the surface energy of the coating. Despite the different contents of siloxane and amphiphilic side chains, the coatings displayed little variation in surface hydrophilicity before hydrolysis (Figure 3), which was $\sim 2^\circ$ difference on WCA between coatings A–F. In contrast, after hydrolysis for 20 days, WCA significantly decreased. Moreover, WCA exhibited an ascending trend in the order of coatings A–F. Through hydrolysis, the surface hydrophilicity of coatings A–F was obviously enhanced, as the reduction value of WCAs ranged from 10 to 18°. Among the nine tested samples, coating A with the highest side chain content of 33% and lowest siloxane content of 28.5% showed a maximum WCA decline of $\sim 18^\circ$, in comparison with the decline of $\sim 4^\circ$ for the three control samples. Considering the molecular structure of the cationic copolymers and the difference in surface hydrophilicity before and after hydrolysis of the nine tested samples, it can be deduced that the cationic PCBMAE moiety may

hydrolyze and yield the zwitterionic structure.^{20,35} Moreover, the side chains promoted the surface hydrophilicity of the coating as the generated zwitterionic groups are capable of binding a layer of water and resisting the adsorption of proteins and microorganisms,¹⁸ whereas the hydrophobic siloxane impaired the surface hydrophilicity. Hence, the higher content of side chains and lower content of siloxane moiety endowed the coating with better hydrophilicity as described above. Of course, the control samples composed of a higher content of siloxane-based main chains definitely have a higher WCA than that of the coatings with side chains.

As for the surface energy, coatings A–F followed a descending order, which was opposite to the WCA. And the control samples have very low surface energy due to the lack of side chains in their molecules. Nonetheless, the zwitterionic side chains are well known for their antibacterial and antifouling characteristics, whereas the low-surface-energy siloxane-based main chains are more fouling-release than fouling-resistance structures.^{7,10} Hence, further exploration of the relationships between structure and fouling-resistance/fouling-release performances is necessary.

3.3. Surface Renewal Speed. Faster surface self-polishing speed correlates with a better fouling-release ability of the

coatings. The weight loss rate reflects the self-renewal ability of the coatings. In this study, both the PCBMAE on the side chain and PCL on the main chain can be cleaved in contact with water. For coatings A–F, the speed of surface renewal increased rapidly in the first 2 weeks until it reached a nearly constant value; oppositely, the control samples maintained a fixed but lower self-renewing speed as shown in Figure 4. Clearly, different contents of side chain and main chain yielded distinctly different surface renewal speeds, because coatings A–C can renew their surfaces faster than that of coatings D–F, and the control samples degraded much more slowly than coatings A–F with amphiphilic side chains. It seemed that the amphiphilic side chains, more specifically, the PCBMAE moiety, dominated the self-replenishment process, followed by the PCL moiety. As shown in Figure 4, coatings A–C with the content of 33% side chains demonstrated significantly faster surface renewal speed than that of coatings D–F with the content of 25% side chains, regardless of the content of PCL. In addition, the surface renewal speed of control 1 with a PCL content of 42.8% was much slower than that of coating F with 16% PCL and 25% side chains. It is well known that the self-replenishment speed is related to fouling-release ability. In this study, the fouling-release ability is decided by the fouling-resistance PCBMAE–PMMA–PVP side chains; that is, the side chains exerted both fouling-resistance and fouling-release functions.

3.4. Surface Morphology. The polarity difference between the cationic copolymers and the coating substrates may easily lead to microphase segregation,²⁶ especially for coatings formulated with a high content of amphiphilic side chains with hydrophilic moieties of PCBMAE and PVP. Here, MMA was introduced to balance the cross-linking effect of the hydrophilic moieties of the side chain. Actually, a maximum quaternization degree of 30% can be achieved for copolymers with weight ratios of side chain to main chain of 1:2 and 1:3. Figure S2 revealed that the coating surfaces, whether fabricated with the synthesized cationic copolymers or just with the siloxane–PCL main chains, were uniform and even before hydrolysis.

When in contact with water, the polymeric fragments resulting from PCL degradation and PCBMAE hydrolysis may accumulate on the surface of the coating; with the erosion of water, they can be washed away. Therefore, the dynamic surface can significantly reduce the contact time and interaction between the organisms and the coating surfaces. By comparing the surface morphology before and after hydrolysis in SEM images, it was found that the hydrolyzed surfaces of coatings A–F were rougher than the prehydrolysis surfaces. Moreover, the hydrolyzed surfaces of coatings A–F were obviously rougher than those of controls, as the side chains facilitated hydrolysis of the cationic copolymer. Nevertheless, the SEM images in Figure S2 showed that there were no obvious cracks or detachment on the coating surface after hydrolysis. This result exhibited the durability of the coatings in an underwater environment, testifying that the copolymers with the monomer components listed in Table 1 and subsequent 30% quaternization were appropriate for coating fabrication.

3.5. Antifouling and Antibacterial Performances. Figure 5 illustrates that the protein adsorption amount on the coating surfaces followed the same trend as that of the surface hydrophilicity, as the generated zwitterionic groups form a hydration layer on the surfaces to prevent the

adsorption of proteins. It is obvious that coatings A–C with a higher content of amphiphilic side chains possess stronger protein resistance ability than that of coatings D–F. Among the tested coatings A–C with 33% amphiphilic side chains, coating A with less siloxane content was more effective in prohibiting protein adsorption than coatings B and C, which agreed well with the coating hydrophilicity illustrated in Figure 3.⁴⁴ Additionally, this result elucidated that the content of amphiphilic side chains is an overwhelming factor in the protein resistance ability. Of course, the self-replenishment of the coating also contributes to the antifouling ability, as the foulant adhered to the surface may peel off together with the polymer fragments. Fortunately, the surface hydrophilicity and surface renewal speed were synergistic, as demonstrated above. Hence, the protein resistance ability of the coating relies more on the amphiphilic side chain rather than on the low-surface-energy main chain. On the contrary, for controls with the molecular structure of siloxane–HDI–PCL, higher contents of siloxane such as control 3 were associated with better antifouling ability due to its lower surface energy. However, even though control 3 has a siloxane content of 64.20%, its protein resistance ability was still not comparable to any of the coatings with an amphiphilic side chain.

Similarly, the antibacterial performance of the coating was also determined by the physicochemical properties on the coating surfaces. For control samples, the antibacterial rate of the copolymer mainly depended on low-surface-energy siloxane, and the antibacterial rate of controls 1–3 was 51–64% against *E. coli* and *S. aureus*. Notably, control 3 with a siloxane content of 64.20% exhibited a significantly higher antibacterial rate than that of control 1 with a siloxane content of 42.80% (Figure 6), despite the fact that control 3 has a slower surface renewal speed.

The antibacterial rate of the coatings prepared from the cationic copolymers followed the ascending order of coatings A–F. The antibacterial rates ranging from 83 to 95% were significantly elevated in comparison with those of the controls, which can be interpreted that the introduction of amphiphilic side chains strengthened the antifouling ability of the coating, as the cations may cause bacterial cell membrane disruption and the zwitterionic structure can inhibit the attachment of bacteria.⁴³ On the other hand, the attachment and settlement of the bacteria can be weakened by siloxane-based materials, and the reduced adhesion force of bacteria on the low-surface-energy siloxane could be more pronounced than the antibacterial efficacy of hydrophilic side chains. Therefore, the coating with suitable hydrophilicity but a higher content of siloxane might be superior to the coating with higher hydrophilicity in terms of antibacterial efficacy. For example, coating F exhibited a higher antibacterial rate than coatings A–E (Figure 6).

3.6. Mechanical Properties. The underwater coating requires good mechanical properties to resist deformation and fracture in response to external forces. In this study, combinations of elastic siloxane and PCL with excellent tensile stress and elongation at break were expected to upgrade the mechanical properties of PDMS-based coating.⁴⁵ Figure 7 shows that control 1 with an equal weight percentage of siloxane and PCL yielded a tensile stress of 9.48 MPa and elongation at break of 678.8%. With increasing weight percentage of siloxane/PCL, its mechanical properties deteriorated. For example, coatings A, B, and C have descending mechanical properties, and the same trend can

be applied to coatings D, E, and F. But for coatings with the same proportion of siloxane and PCL, such as coatings A and D, coatings B and E, and coatings C and F, coatings D, E, and F possessed better mechanical properties than their counterparts coatings A, B, and C, respectively (Figure 7). The above results indicated that the introduction of amphiphilic side chains damaged the integrity of the polymer structure, diminishing its mechanical properties. Also, longer amphiphilic side chains could cause a more disordered structure of the coating, undermining its ability to withstand the external forces. Considering the previous reports on siloxane-based marine coatings,⁴⁶ the mechanical properties of the coatings are able to meet the criteria of underwater coating.

3.7. Marine Field Test. Marine field tests were conducted during the thriving growth season of marine organisms from May 15 to Oct 12. The pictures in Table 2 illustrate that coating F was able to prohibit the attachment of marine organisms for 4 months, while the algae started to grow on the commercially available CSPC after at least 3 months, probably owing to the excessive amount of environmentally toxic Cu₂O in the coating. Several spots on the surface of coatings B and C were attached by a few hard foulants like barnacles within 3 months. But with the growth of the marine organisms, most of the surface areas were occupied by foulants in 5 months. In contrast, coatings A, D, and E showed less favorable antifouling results, probably due to their low content of siloxane. The control samples, especially control 1, were completely taken by marine organisms in 2 months; in contrast, coatings C and F with a similar siloxane content as that of control 1 presented much stronger antifouling capabilities. When increasing the siloxane content to 57.06% as in control 2 and 64.20% as in control 3, the antifouling performances of the controls were greatly upgraded in the time duration of 2 months. Nevertheless, the antifouling abilities of the control samples were not as strong as those of coatings B, C, and F. These results indicated that the introduction of the amphiphilic side chains indeed promoted the antifouling ability of the coating, and a higher content of fouling-release siloxane is crucially important to achieve better antifouling performances. Overall, coating F with 25% amphiphilic side chain, 48% siloxane, and 16% PCL was the most effective in marine field test.

4. CONCLUSIONS

Cationic polymers with random copolymerized PCBMAE–PMMA–PVP as the side chain and siloxane and PCL alternatively arranged via HDI as the main chain were synthesized via a one-pot continuous reaction to prepare marine antifouling coatings. The experimental results manifested that the introduction of hydrolyzable side chains enhanced the surface hydrophilicity, protein resistance, and antibacterial rate of the coating compared with the control coatings prepared with copolymers only having a siloxane–PCL main chain. The fouling-release activity of the coating was regulated via surface self-renewal and the low-surface-energy siloxane moiety. Experimental results illustrated that, other than the surface renewal speed, the siloxane content of the cationic copolymers is more important to achieving better antibacterial performance. This study explored the mechanisms of fouling release and fouling resistance through the combination of multifunctional domains, providing ideas for the synthesis of new environmentally friendly materials to solve the marine fouling problem.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.5c00179>.

Expanded ¹H NMR spectra of copolymers (Figure S1 in Section 3.1) and SEM images of coatings before and after hydrolysis (Figure S2 in Section 3.4) (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors are grateful for the financial support from the National Natural Science Foundation of China (No. U2106226).

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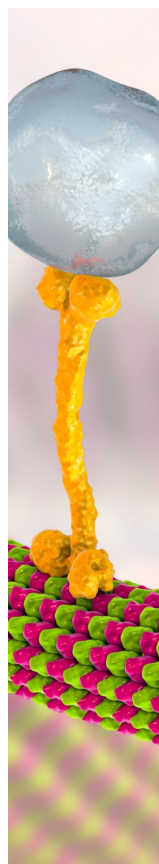
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